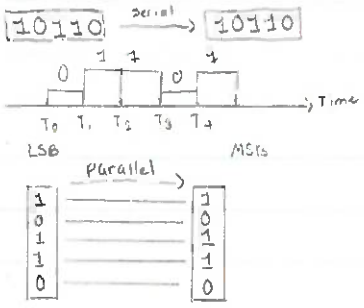
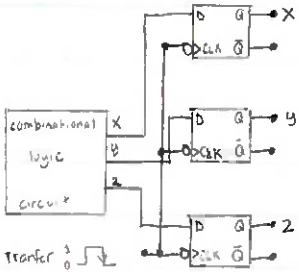
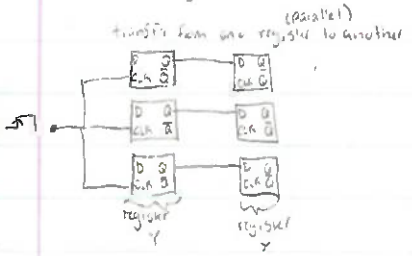
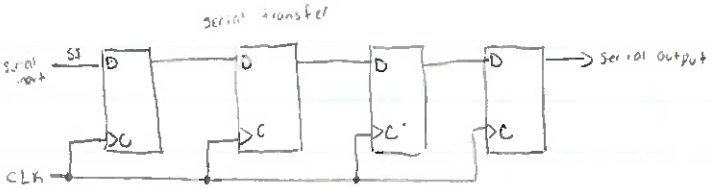
Chapter 6 - Registers & counters

- Multiple flip-flops can be combined to form a data register
- Shift registers allow data to be transported one bit at a time
- Registers also allow for parallel transfer (many bits transfered at the same time)





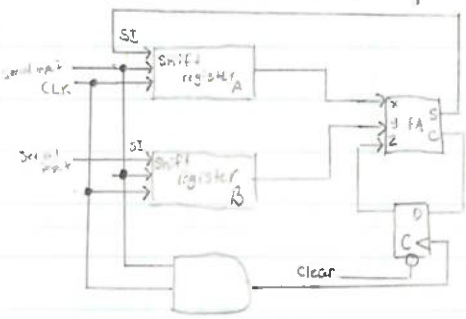




Serial transfer

Time	A	B
T0	1011	0011
T1	1101	1001
T2	1110	1100
T3	0111	0110
T4	1011	1011

Serial addition (D Flip Flop)



Sequential circuit: design procedure

1. From the word description and specifications of the desired operation, derive a state diagram for the circuit.
2. Obtain the state table
3. Reduce the number of states if necessary
4. Assign binary values to the states
5. Determine the number and types of flip-flops to be used
6. Obtain simplified flip-flop input equations and output equations
7. Draw the logic diagram

Excitation table JK Flip-Flop

J-K Flip-Flop			
Present	Next	Input J	Input K
0	→ 0	0	X
0	→ 1	1	X
1	→ 0	X	1
1	→ 1	X	0